Supplementary Material for "*Ab initio* modeling and experimental investigation of Fe_2P by DFT and spin spectroscopies"

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A. NMR

The NMR spectra were reconstructed point by point by exciting spin echoes at discrete frequencies. The corresponding spectral amplitude were determined as the zero-shift Fourier component of the echo signal, divided by the frequency-dependent sensitivity $\propto \omega^2$. At each frequency step, the LC resonator was re-tuned by a servo-assisted automatic system plugged into the spectrometer. A standard $P - \tau - P$ spin echoes pulse sequence was employed, with equal rf pulses P of intensity and duration optimized for maximum signal, and delay τ as short as possible, compatibly with the dead time of the apparatus.

B. DFT simulation details

The magnetic ground state of Fe_2P is accurately reproduced by the Perdew-Burke-Ernzerhof (PBE) exchange and correlation functional [S1]. The energy difference between the ferromagnetic and the spin-degenerate case is 1.1 eV per unit cell and the magnetic moment of Fe1 is found to be 0.84 μ_B while that of Fe2 is 2.22 μ_B. Both these results are in agreement with previous computational studies and compare extremely well with experimental findings.[S2–S5]

The lattice structure of Fe_2P was taken from the crystallographic information file deposited on the COD database [S6] with ID: 9012616.

In the simulations using plane wave and pseudopotential, the basis set was expanded up to a cutoff of 80 Ry while charge density up to 800 Ry. A uniform Monkhorst-Pack (MP) [S7] mesh of $6 \times 6 \times 8$ was used for sampling the reciprocal space in the unit cell while we used a $2a \times 2b \times 3c$ supercell to identify muon interstitial sites.

In the optimization of the structure and description of magnetic properties a Marzari-Vanderbilt [S8] smearing of 5 mRy was used while a Gaussian smearing of 20 mRy is chosen for supercell calculations.

The experimental lattice parameters $a = b = 5.877$ and $c = 3.437$Å [S9–S11] have been used throughout all the calculations. The optimized atomic positions show very small displacements , always smaller than $5 \cdot 10^{-2}$ Å. In order to further validate our description we also optimized lattice parameters and found $a = b = 5.815$ and $c = 3.425$ Å.

In all cases we performed spin-polarized calculations using the generalized gradient approximation (GGA) proposed by Perdew Burke and Ernzerhof (PBE)[S1] and Projector Augmented Wave (PAW) [S12] pseudopotentials that are required to reconstruct the core electrons polarization.

For muon site assignment, the set of initial muon locations and the Fe_2P atomic positions were fully relaxed in a $2a \times 2b \times 3c$ supercell until forces and energy difference between optimization steps were smaller than 10^{-3} Ry/a.u and 10^{-4} Ry. The full list of positions used to perform the calculation is in Tab. I.

A $4 \times 4 \times 4$ uniform grid is used to sample the interstitial voids in the unit cell. The number points actually used as starting guess for the muon site reduces to 8 when positions closer than 1 Å to hosting atoms and symmetry equivalent points have been removed. Three additional positions were identified from the disconnected minima of the bulk electrostatic potential and used as an additional starting guess. Finally, for the estimation of the contact hyperfine field at the muon sites, the reciprocal space sampling was increased to a $4 \times 4 \times 4$ Monkhorst-pack mesh grid.

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TABLE I. Summary of candidate muon stopping sites. The first 8 sites are from $4 \times 4 \times 4$ uniform grid and the last 3 from the bulk electrostatic potential.

	Wyckoff[a]	(x, y, z)[b]	(x, y, z)[c]	$\Delta E(eV)$[d]
1	$1a$	0.00 0.00 0.00	0.0000 0.0001 0.0001	1.1671
2	$3g$	0.00 0.25 0.50	0.0024 0.3274 0.5000	0.0006
3	$3f$	0.00 0.50 0.00	0.0000 0.5473 0.0074	0.7464
4	$6i$	0.00 0.50 0.25	0.0002 0.5314 0.1038	0.7344
5	$3f$	0.00 0.75 0.00	0.0000 0.7068 0.0003	0.2545
6	$6i$	0.00 0.75 0.25	0.0000 0.7054 0.0160	0.2556
7	$12l$	0.25 0.50 0.25	0.0578 0.3595 0.4997	0.0009
8	$6k$	0.25 0.50 0.50	0.0486 0.3535 0.5000	0.0000
9	$6k$	0.10 0.35 0.50	0.0554 0.3572 0.4999	0.0003
10	$3f$	0.35 0.35 0.00	0.2959 0.2959 0.0038	0.2549
11	$2d$	0.33 0.66 0.50	0.3320 0.6654 0.5001	0.6324

[a] wyckoff number
[b] starting position
[c] final relaxed position
[d] total DFT energy difference w.r.t lower energy site

Code	Nucleus	Cont.	Dipolar SR	Orbital SR
Elk	P1	4.0(5)	-0.8	0.1
	P2	0.2(5)	0.1	0.1
	Fe1	-15.8(5)	0.6	1.5
	Fe2	-17.2(6)	-0.9	4.6
Wien2K	P1	3.8	-1.0	0.0
	P2	0.3	0.2	0.0
	Fe1	-13.1	0.5	1.5
	Fe2	-14.9	-0.7	4.3

TABLE II. Hyperfine field at the P and Fe nuclei obtained with Elk and Wien2K with magnetic moments parallel to the **c** lattice vector. All values are in Tesla and are obtained from simulations including spin orbit interaction. The numbers appearing in the brackets are the standard deviation of the two estimates of the contact part obtained with the two different methods implemented in Elk.

For all electron simulations performed with Elk, in order to achieve convergence of hyperfine fields estimates, a dense $10 \times 10 \times 12$ MP reciprocal space grid is used while atomic positions were kept fixed in the experimentally reported values. For simulations with spin-orbit contribution, the k-point grid was reduced to $7 \times 7 \times 9$, to speedup the simulation at a negligible loss of accuracy on the scale of the uncertainties discussed below. A plane-wave cutoff of $|G + K|_{\max} = 9/R_{\min}^{MT}$ ($R_{\min}^{MT}$ is the average of the muffin-tin radii in the unit cell) was used for the expansion of the wavefunction in the interstitial region. The muffin-tin radius for Fe and P are 1.98 and 2.17 a.u. and respectively. The number of empty states was increased until it reached 40% of valence states and spin-orbit coupling was considered. For Fe, we moved semi-core s states to the core and treated them with the full Dirac equation including core polarization.

The results from WIEN2k code were obtained with version 18.2. In this case the reciprocal space grid was $12 \times 12 \times 18$ and muffin-tin radii were 2.35 and 1.83 a.u. for Fe and P respectively. The plane waves were expanded up $R_{MT}k_{max} = 8$ where R_{MT} is the average radius of MT spheres while non-spherical potential and charge density inside MT were expanded up to $l = 10$. A total charge difference smaller than $10^{-6}e$ has been set as SCF convergence criterion.

C. Hyperfine Fields

In magnetic materials, both core and valence electrons contribute to the hyperfine field, with the former requiring the solution of the Dirac equation to correctly account for relativistic effects in heavy nuclei. The hyperfine field in localized magnetic system can be split into a contribution from the electronic density surrounding the spin of interest within a typical atomic dimension (short-range contribution) and a contribution from electron density localized at distant sites (long-range contribution). The latter can be described in the classical limit by treating the distant spin polarized electronic orbitals as classical magnetic dipoles.

Let us focus on the short-range interaction first that reads, for the electron with spin **S** and the nucleus or the muon with spin $\mathbf{I}_n$,

$$\mathcal{H}^{HF} = \mathcal{H}_C + \mathcal{H}_D + \mathcal{H}_L \tag{1}$$

$$\mathcal{H}_C = \frac{\mu_0}{4\pi}\frac{8\pi}{3}\gamma_e\gamma_{j_n}\delta\left(\mathbf{r}_n\right)\mathbf{S}\cdot\mathbf{I}_n \tag{2}$$

$$\mathcal{H}_D = \frac{\mu_0}{4\pi}\gamma_e\gamma_{j_n}\frac{3\left(\mathbf{e}_n\cdot\mathbf{S}\right)\left(\mathbf{e}_n\cdot\mathbf{I}_n\right)-\mathbf{S}\cdot\mathbf{I}_n}{r_n^3}, \tag{3}$$

$$\mathcal{H}_L = \frac{\mu_0}{4\pi}\gamma_e\gamma_{j_n}\frac{\mathbf{L}_n\cdot\mathbf{I}_n}{r_n^3} \tag{4}$$

where μ_0 is the vacuum permeability, γ_e is the electron gyromagnetic ratio and γ_{j_n} is the j-nth nuclear or muon gyromagnetic ratio, $\mathbf{r}_n = \mathbf{r} - \mathbf{R}_n$ is the electrons position relative to the j-nth nucleus, $r_n = |\mathbf{r}_n|$ and $\mathbf{e}_n = \mathbf{r}_n/r_n$ and $\mathbf{L}_n = \mathbf{r}_n \times \mathbf{p}$.

The contributions in Eq. 1 are estimated using different approaches in WIEN2k and Elk. WIEN2k exploits the fact that inside the muffin-tin the basis functions are atomic-like and the expectation values of operators acting on electrons space in Eq. 1 can be easily computed if the interstitial part is neglected. The contact part is finally estimated with an average over the Thomson radius [S13] of the spin density at the nucleus to account for relativistic corrections. The Elk code computes instead the total field at nuclear sites by solving the Poisson equation for the vector potential where the current density is obtained from the electronic core and valence spin magnetization density and, optionally, from the orbital current density. The value is eventually averaged over the nuclear radius [S14] since core electrons' contribution is obtained from the Dirac equation. When the orbital contribution is neglected, the isotropic part of the hyperfine tensor obtained with this latter method represents an alternative estimate of the contact hyperfine field that is also directly obtained from the average of the electronic spin density over the nuclear radius. The results obtained with these two methods can be used to quantify the accuracy of the computational estimate and is found to be larger than the convergence obtained against the basis set. The standard deviation of the values obtained with the two approaches is reported in Tab. II.

Finally, the long range interaction is obtained with the approach proposed by Lorentz which involves the sum of two contributions: a real space sum of the field produced by the magnetic dipoles inside a large sphere of radius r_L and a second contribution due to magnetic moments outside the sphere. These simulations have been converged against the size of r_L and a $100 \times 100 \times 200$ supercell was constructed to host the sphere.

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